

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 16****Topics covered :**

1. Method of undetermined coefficients

- The annihilators (denoted as $Q(D)$) for some common functions are given in Table 1

Table 1: Annihilators for some common function

$g(x)$	Annihilator $Q(D)$
$\sum_{i=0}^n a_i x^i$	D^{n+1}
$\sum_{i=0}^n a_i x^i e^{ax}$	$(D - a)^{n+1}$
$\sum_{i=0}^n a_i x^i e^{ax} \cos(mx)$	$\left((D - a)^2 + m^2\right)^{n+1}$

- When applying the annihilator $Q(D)$ to both sides of the non-homogeneous ODE $L[y] = g(x)$ (or equivalently $p(D)y = g(x)$), we obtain

$$Q(D)\{P(D)y\} = Q(D)g(x) = 0 \quad (1)$$

For differential equations with constant coefficients, (1) can be rewritten as

$$\underbrace{Q(D)P(D)}_{R(D)} y = R(D)y = 0 \quad (2)$$

The core idea behind the [Method of undetermined coefficients](#) is to find the particular solution of $L[y] = g(x)$ by using the solution to the associated homogeneous differential equation $R(D)y = 0$.

If $P(D)$ is a polynomial of degree n and $Q(D)$ is a polynomial in D of degree m , then, $R(D)$ is a polynomial of degree $n + m$. As a result, when applying the annihilator of order m to the non-homogeneous ODE $L[y] = g(x)$ results in a homogeneous ODE $Q(D)P(D)y = 0$, which is of the order $m + n$.

- The general solution of a homogeneous constant coefficient ODE of order n depends on the nature of the roots of the auxiliary equation

1. Real and distinct roots: When the roots of the auxiliary equation (denoted by s_i) are real and distinct, the general solution is given by

$$y(x) = \sum_{i=1}^n c_i y_i(x) \text{ where } y_i(x) = e^{s_i x} \quad (3)$$

2. Real and repeated roots: If any root of the auxiliary equation is repeated (say m times), the general solution is of the form

$$y(x) = \sum_{i=1}^{n-m-1} c_i y_i(x) + \sum_{i=n-m}^n c_i y_i(x) \quad (4)$$

where

$$y_i(x) = e^{s_i x} \quad \forall i \in \{1, \dots, n-m-1\} \quad \text{and} \quad y_i(x) = x^{i-n-m} e^{s_m x} \quad \forall i \geq n-m \quad (5)$$

It is essential to apply this logic correctly in the case of repeated roots, even if the multiplicity of the root varies.

3. Complex and repeated roots: If any of the complex roots (e.g., $-\sigma + j\omega_d$) are repeated m times, then the general solution corresponding to the repeated roots is of the form

$$e^{-\sigma x} \left[(c_0 + c_1 x + c_2 x^2 + \dots + c_{m-1} x^{m-1}) \cos(\omega_d x) + (c_0 + c_1 x + c_2 x^2 + \dots + c_{m-1} x^{m-1}) \sin(\omega_d x) \right] \quad (6)$$

It is important to note that the general solution also includes terms corresponding to the other distinct roots, which must be incorporated.

- **Method of Undetermined Coefficients**: The core idea of the Method of Undetermined Coefficients is to determine the particular solution of the non-homogeneous ODE $L[y] = g(x)$ by using the solution of the corresponding homogeneous ODE. This is achieved by applying the annihilator to both sides, transforming the problem into solving $R(D)y = 0$, where $R(D) = P(D)Q(D)$.

The roots of the auxiliary equation of $R(D)$ will include the roots of the auxiliary equation corresponding to $P(D)$. Assuming the roots of the auxiliary equation of $R(D)$ are arranged such that the roots of $P(D)$ appear first, the general solution of the homogeneous equation $R(D)y = 0$ can be expressed as follows:

$$y = \underbrace{\sum_{i=1}^n c_i y_i(x)}_{y_c(x) \text{ for } P(D)y=0} + \underbrace{\sum_{i=n+1}^{n+m} c_i y_i(x)}_{\text{Term I}} \quad (7)$$

From Lecture 15, recall that the general solution of the non-homogeneous ODE $L[y] = g(x)$ is given by $y(x) = y_c(x) + y_p(x)$. Referring to Equation 7, it can be inferred that if the coefficients c_i for $i > n$ can be determined such that $P(D)y = g(x)$ (or more precisely, $P(D)(\text{Term I}) = 0$, since $P(D)y_c = 0$), then Equation (7) represents the general solution of $L[y] = g(x)$.

Thus, the steps involved in the method of undetermined coefficients are:

1. **Determine the annihilator and compute the homogeneous ODE**: Determine the annihilator for $g(x)$. Apply the annihilator $Q(D)$ to both sides of the non-homogeneous equation $L[y] = g(x)$. This transforms the non-homogeneous equation into a homogeneous one $R(D)y = 0$.
2. **Find the complementary solution**: Solve the corresponding homogeneous ODE $R(D)y = 0$. Identify the parts of the solution that correspond to the complementary solution $y_c(x)$ for $P(D) = 0$ (usually the first n terms).
3. **Find the particular solution**: For the remaining terms in the general solution of $R(D)y = 0$ (i.e., terms that are not a part of the complementary solution for $P(D) = 0$ i.e., from $n+1$), determine their coefficients c_i such that the linear combination of these terms satisfy $P(D) \sum_{i=n+1}^{n+m} c_i y_i(x) = g(x)$. The particular solution is thus $y_p(x) = \sum_{i=n+1}^{n+m} c_i y_i(x)$

4. **Write the general solution:** The general solution to the non-homogeneous ODE is then the sum of the complementary solution $y_c(x)$ and the particular solution $y_p(x)$, i.e., $y(x) = y_c(x) + y_p(x)$.